

WSPRadar.org reconstruction & comparison

SNR difference by time of day

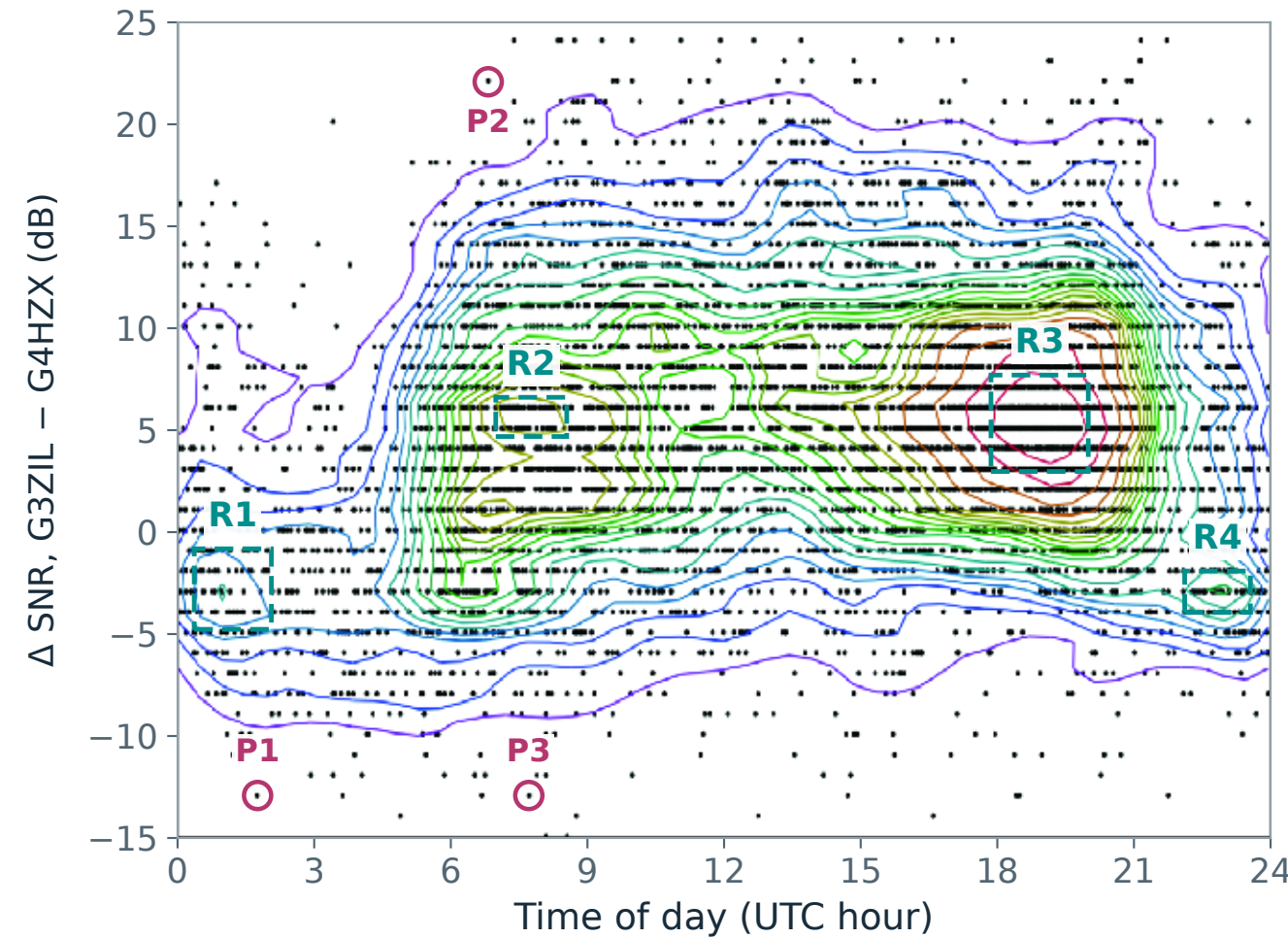
Referenced publication: Gwyn Griffiths (G3ZIL) and Nigel Squibb (G4HZX) (2017). *Improving HF Band SNR*.

Source figure: **Figure 6, p. 25** · Practical Wireless, October 2017, pp. 23-26

Demo: 5-7 April 2017 | 40 m | $\Delta\text{SNR} = \text{G3ZIL} - \text{G4HZX}$ | B/C: 6,459 paired observations

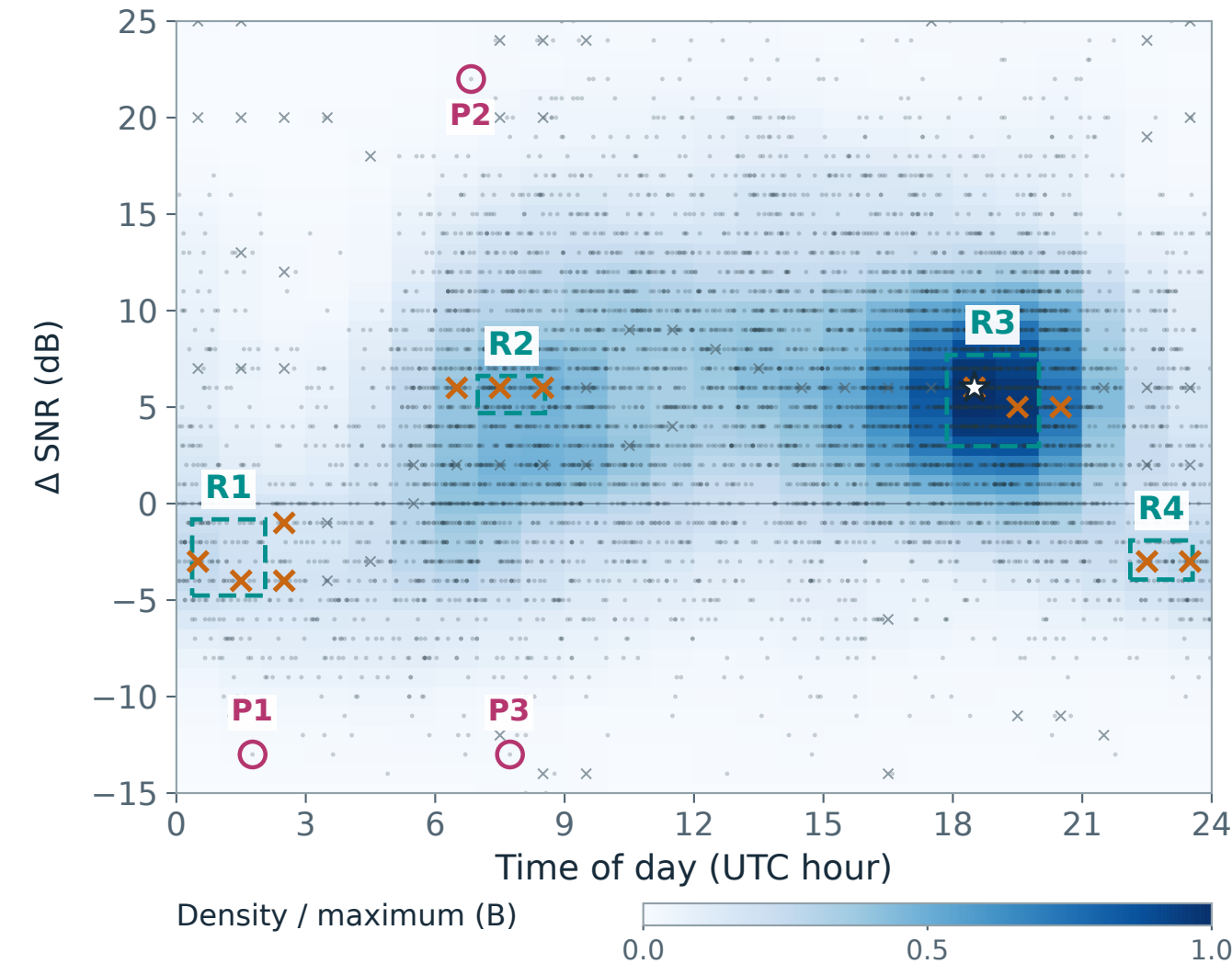
A Original image from publication

Original scatter and contours · linear dB



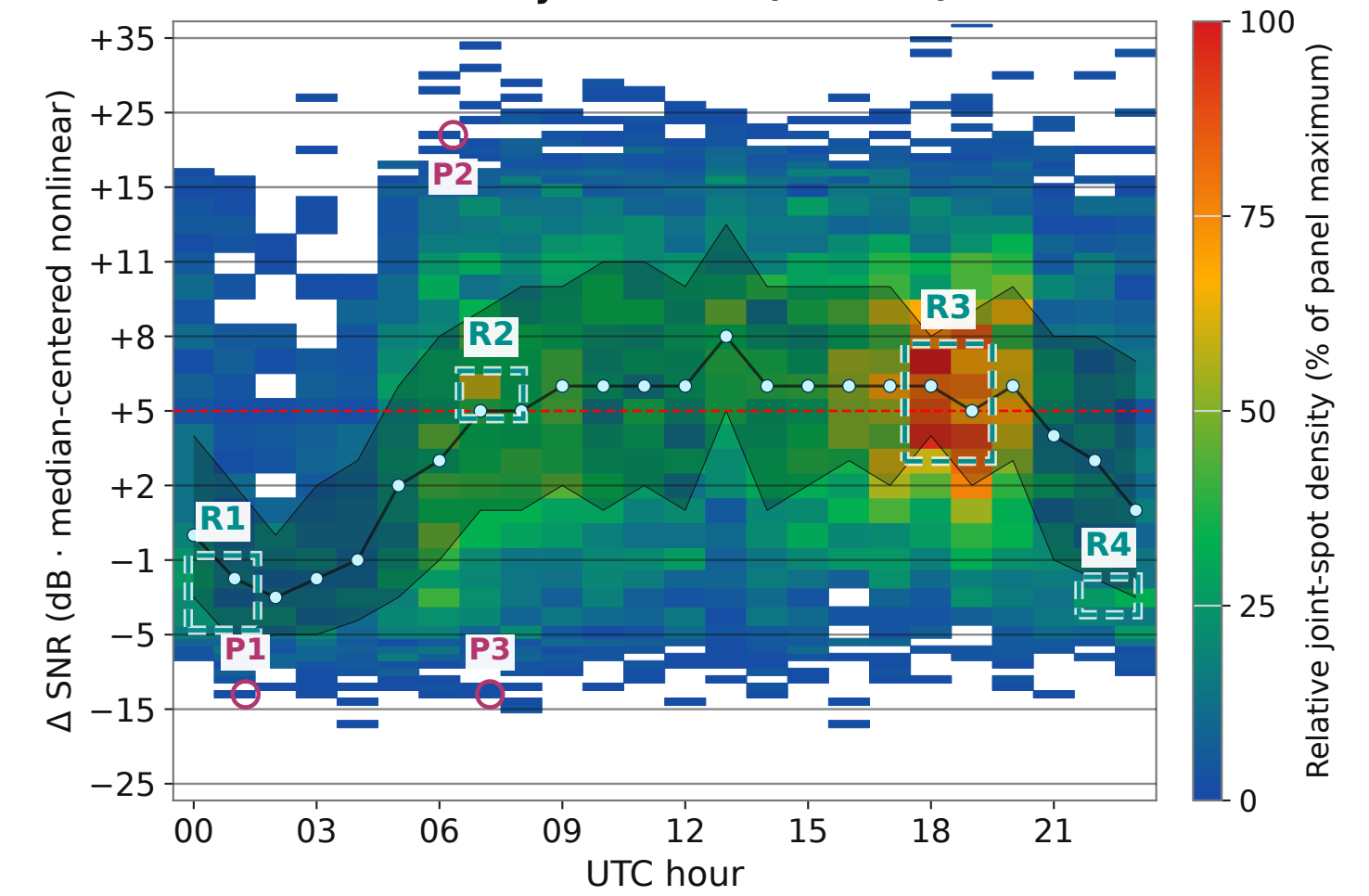
B Reconstruction

Smoothed observation density · linear dB



C WSPRadar view

ΔSNR by UTC Hour (1 h bins)



- R1-R4: paper contour regions (A-C)
- P1-P3: selected point matches (A-C)
- Individual paired observations (B)

- × Matching density peaks (B)
- × Other local density peaks (B)
- ☆ Highest reconstructed density (B)

- Overall median: +5 dB (C)
- Median for each UTC hour (C)
- Middle 50% for each UTC hour (C)

R1 Early-night negative branch

R2 Morning positive branch

R3 Strongest evening island

R4 Late-night negative branch

Panel C: Same observations, native WSPRadar presentation. R1-R4 map the same paper bounds onto the app's axis. P1-P3 mark matched native spots within hourly cells. Unsmoothed 1-hour × 1-dB cells; pooled median +5 dB. Markers: hourly medians. Band: hourly middle 50% (IQR). C uses the app's median-centered nonlinear dB axis. Read the dB labels; vertical pixel positions differ from A/B. Medians and IQRs use freshly recalculated pairs; they are not measurements extracted from the paper.

A/B alignment: printed-axis calibration, with no fitted shift or scale. R1-R4 are contour regions, not confidence intervals.

Matching peaks use the fixed ± 4 -pixel readout + half-cell allowances; boxes show the source regions alone. P1-P3 match folded time/SNR, not identified dates or stations.

B: database reports → WSPRadar SQL and paired evidence → 1-hour × 1-dB grid → fixed comparison smoother ($\sigma = 1$ hour, 1 dB; UTC wraps).

A/B display −15 to +25 dB. Faint points are native pairs; gray × mark other density peaks. C retains the full app view.

Original figure: p. 25, Figure 6. Selection: 5 April 00:00-7 April 23:45 UTC; distance <10,000 km.

WSPRadar.org by Dr. Markus Brosch (DL1MKS)